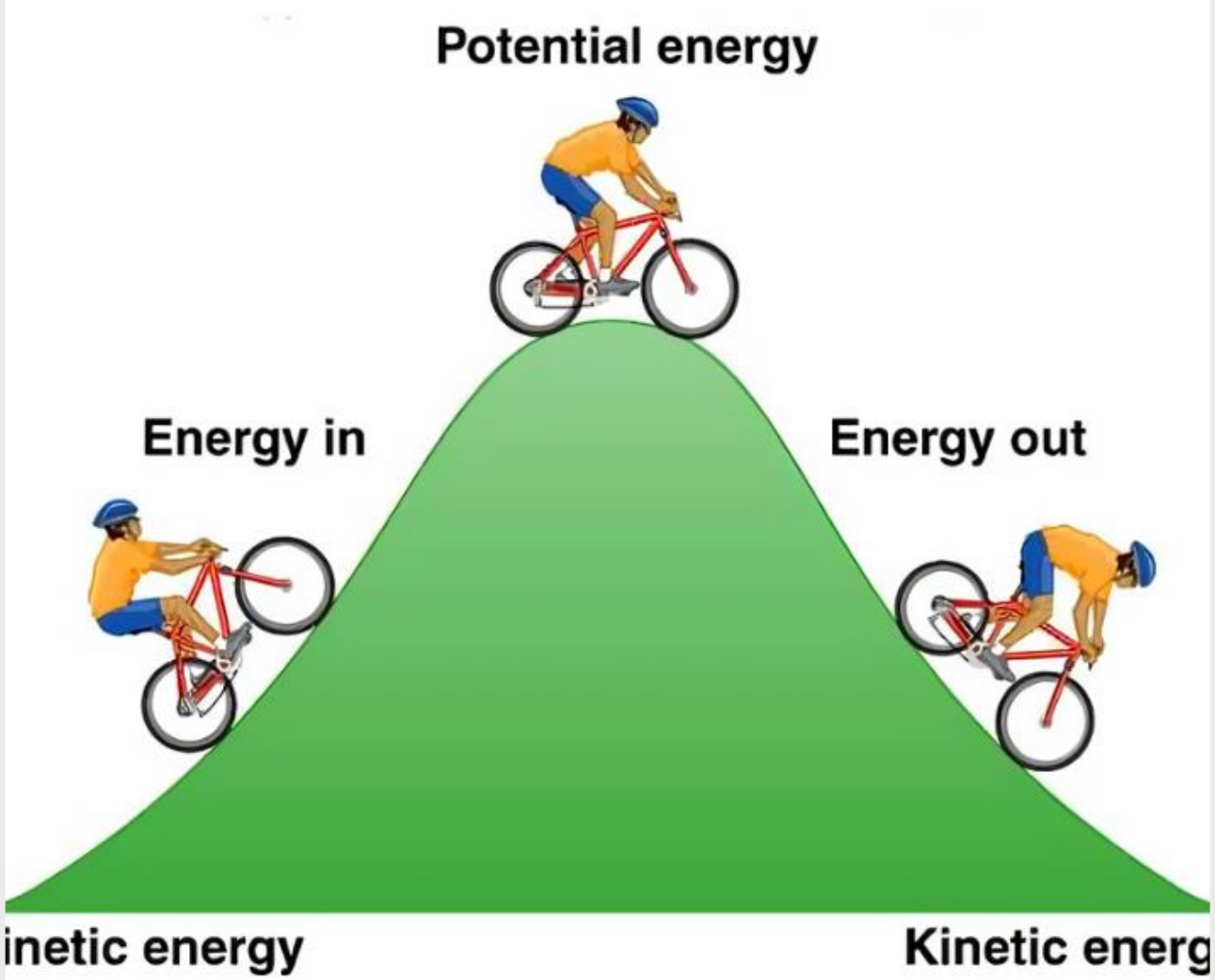
Chapter 8

CONSERVATION OF ENERGY

Định luật bảo toàn và chuyển hóa năng lượng



NỘI DUNG CHÍNH

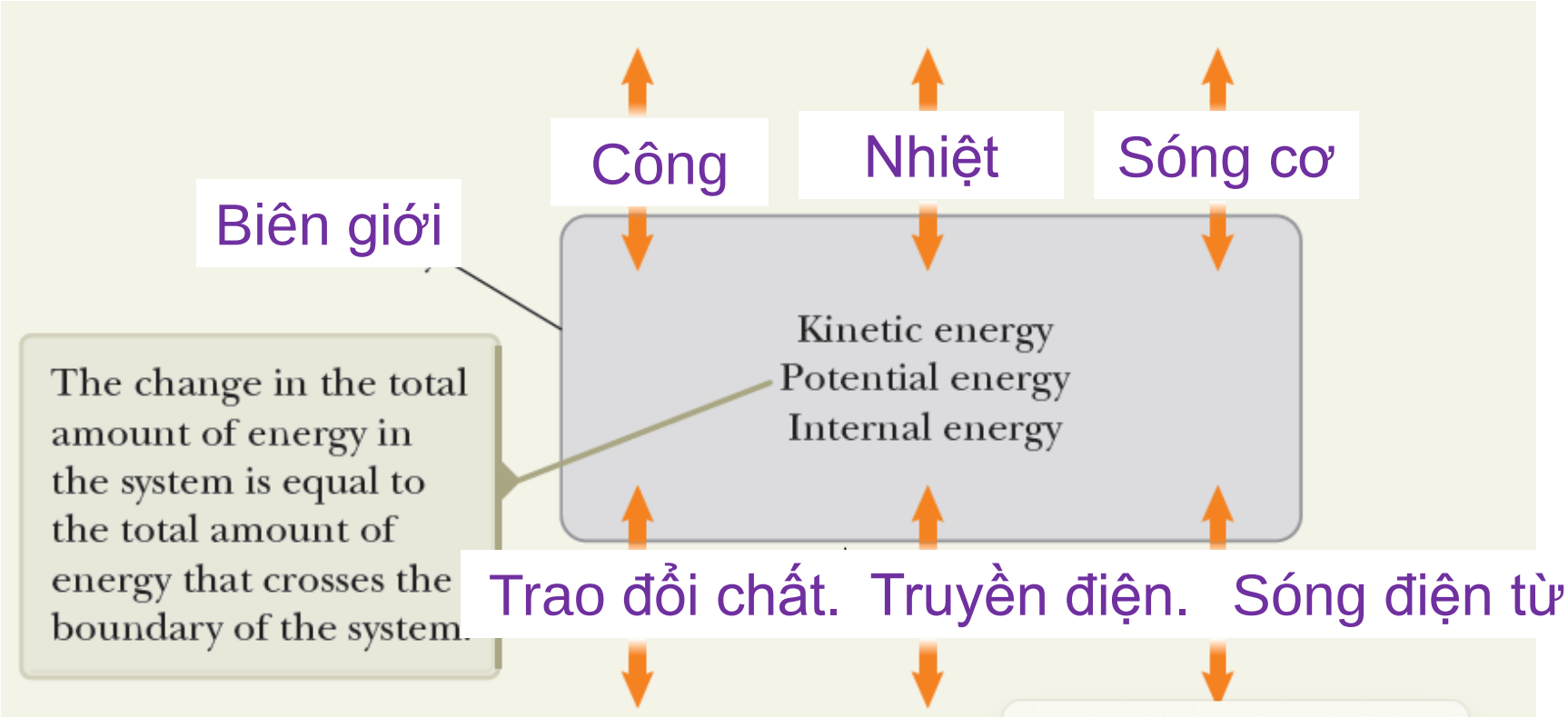
- Hệ không cô lập
- Hệ cô lập
- Các định luật bảo toàn năng lương đối với hệ cô lập và hệ không cô lập
- Bài tập ví dụ
- Công suất

HỆ KHÔNG CÔ LẬP

- Hệ không cô lập về năng lượng là một hệ có trao đổi năng lượng với môi trường qua biên giới của nó
- Một hệ không cô lập sẽ tương tác với môi trường

HỆ KHÔNG CÔ LẬP

- Đối với hệ không cô lập: năng lượng của hệ được trao đổi với môi trường bên ngoài trong thời gian hệ tương tác với môi trường.
- Năng lượng của hệ được trao đổi với môi trường bên ngoài theo các cơ chế sau:



HỆ KHÔNG CÔ LẬP

Năng lượng được truyền vào hệ dưới dạng công

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a

Năng lượng được truyền đi dưới dạng sóng cơ

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b

Năng lượng mất mát dưới dạng nhiệt

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c

Năng lượng được truyền vào hệ dưới dạng trao đổi chất

pcoon/Photodisc/Getty Images

Năng lượng được truyền vào hệ bằng truyền điện

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Năng lượng ra khỏi hệ dưới dạng sóng điện từ

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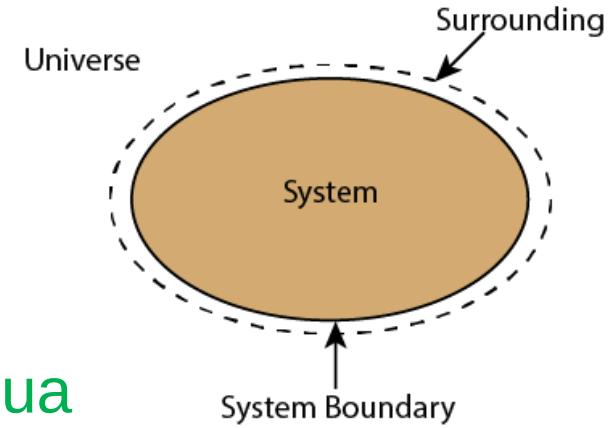
HỆ KHÔNG CÔ LẬP

Phương trình bảo toàn năng lượng:

$$\Delta E_{system} = \sum T$$

Năng lượng toàn phần

Năng lượng được truyền qua biên giới của hệ



$$\Delta K + \Delta U + \Delta E_{int} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

Với W: công, Q: nhiệt,
K: động năng, U: thế năng và E_{int}: nội năng
T_{MW}: sóng cơ,
T_{MT}: trao đổi chất,
T_{ET}: truyền điện,
T_{ER}: sóng điện từ

HỆ KHÔNG CÔ LẬP

Bảo toàn năng lượng:

$$\Delta K + \Delta U + \Delta E_{int} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

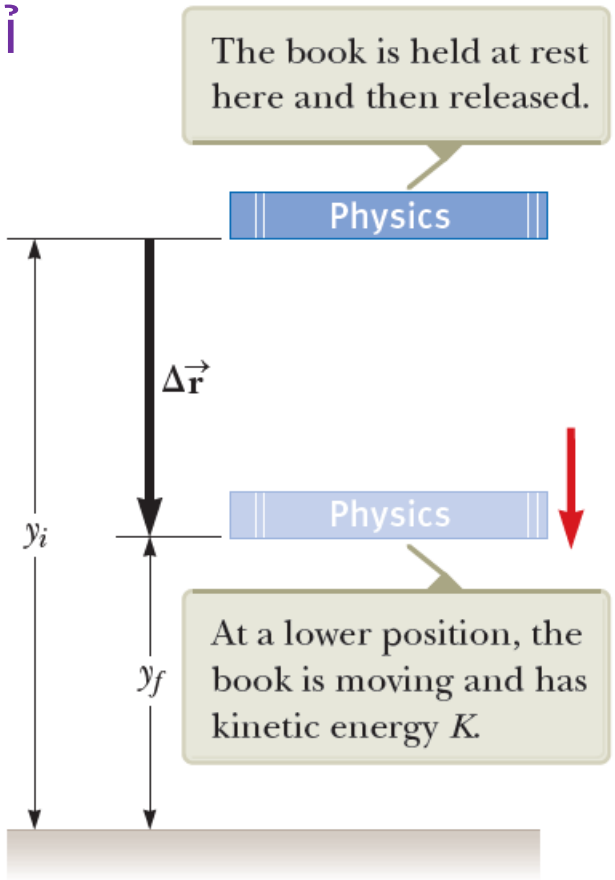
Giả sử công gây ra sự thay đổi về động năng của một hệ:

$$\Delta K = W$$

Định lý công-động năng

HỆ KHÔNG CÔ LẬP

Xét hệ là cuốn sách (hệ không cô lập): rơi từ trạng thái nghỉ dưới tác dụng của trọng lực



Công thực hiện bởi trọng lực lên cuốn sách:

$$W_{on\ book} = \vec{F}_g \Delta\vec{r} = mg|\Delta\vec{r}|\cos 0^o = mg(y_i - y_f)$$

Đinh lý công-động năng

$$W_{\text{on book}} = \Delta K_{\text{book}}$$

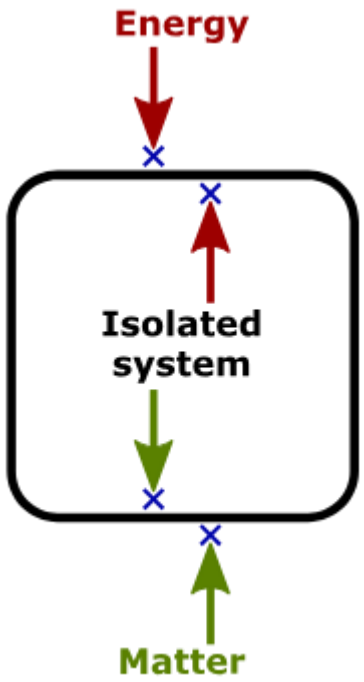
Công thực hiện lên sách Biến thiên động năng

$$\Delta K_{\text{book}} = mgy_i - mgy_f$$

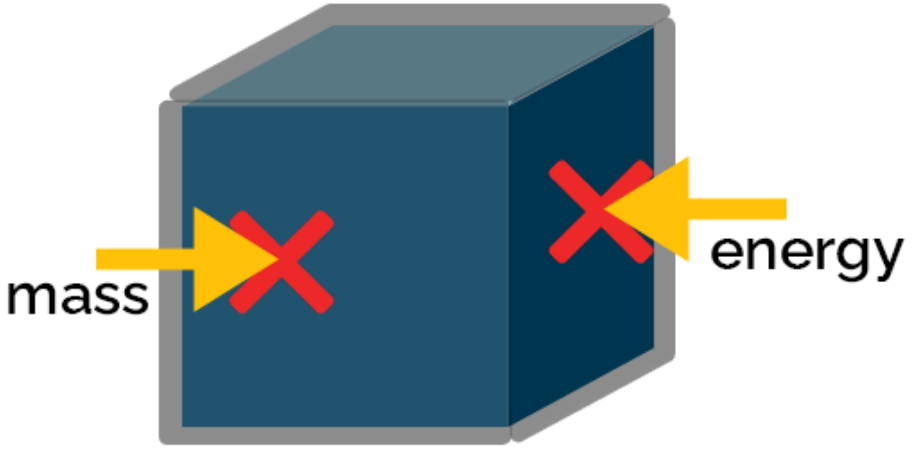
HỆ CÔ LẬP

Hệ cô lập: là một hệ không trao đổi năng lượng với môi trường qua biên giới của hệ.

HỆ CÔ LẬP



a physical system through which neither mass nor energy can pass



isolated system

Game Smartz flashcard

https://gamesmartz.com/definitions?definition=8682&isolated-system&s=85

Hệ cô lập: Bảo toàn năng lượng

$$E_{system} = K + U + E_{int} = \text{const}\ (hệ\ cô\ lập)$$

Năng lượng của hệ = Động năng + Thế năng + Nội năng

$$\boxed{\leftrightarrow \Delta E_{system} = \Delta K + \Delta U + \Delta E_{int} = 0\ (hệ\ cô\ lập)}$$

Độ biến thiên năng lượng của hệ = Độ biến thiên Động năng + Độ biến thiên Thế năng + Độ biến thiên Nội năng

Không có lực ma sát ➔ *Nội năng*: $\Delta E_{int} = 0$

$$\boxed{\Delta E_{mech} = \Delta K + \Delta U = 0 \leftrightarrow E_{mech} = K + U = const}$$

Cơ năng trong hệ cô lập sẽ bảo toàn khi không có lực không bảo toàn (ví dụ ma sát, lực cản) tác dụng

Khi có lực ma sát động: $\Delta E_{int} = \mathrm{f_k.d}$

$$\boxed{\Delta K + \Delta U + \mathrm{f_k.d} = 0}$$

Sự thay đổi nội năng do lực ma sát động gây ra

HỆ CÔ LẬP

Xét hệ gồm sách và trái đất: hệ cô lập

=> $\Delta E_{system} = 0$

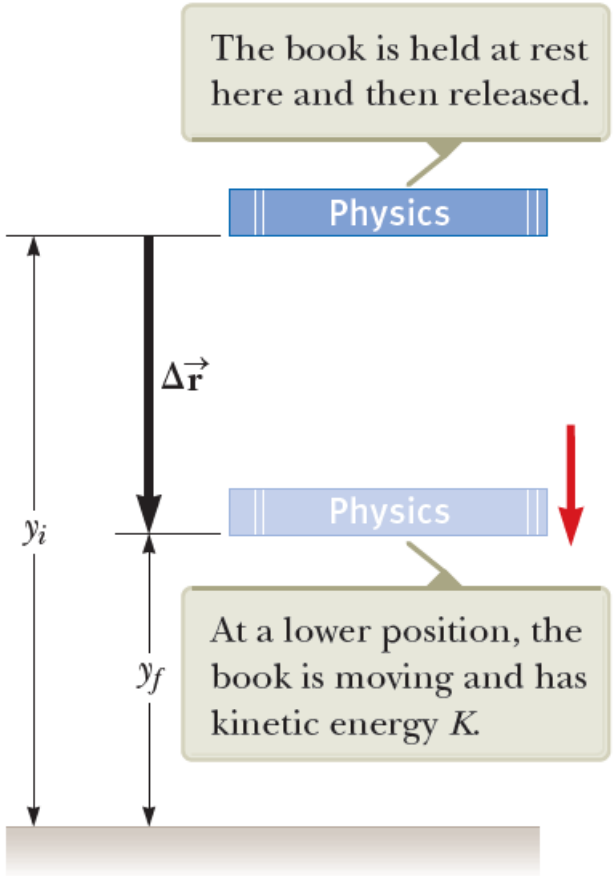
=> $\Delta K_{book} + \Delta U_g = 0$

=> $\Delta K_{book} = -\Delta U_g = -(mgy_f - mgy_i)$

$$\frac{1}{2}mv_f^2 + mgy_f = \frac{1}{2}mv_i^2 + mgy_i$$

=> Bảo toàn cơ năng:

$$\boxed{\Delta E_{mech} = \Delta K + \Delta U = 0}$$



Cơ năng: $E_{mech} \equiv K + U$

Động năng

Thế năng

HỆ CÔ LẬP

Nếu có lực không bảo toàn (ví dụ lực ma sát) tác dụng lên hệ cô lập
=> Năng lượng toàn phần của hệ bảo toàn nhưng cơ năng không bảo toàn:

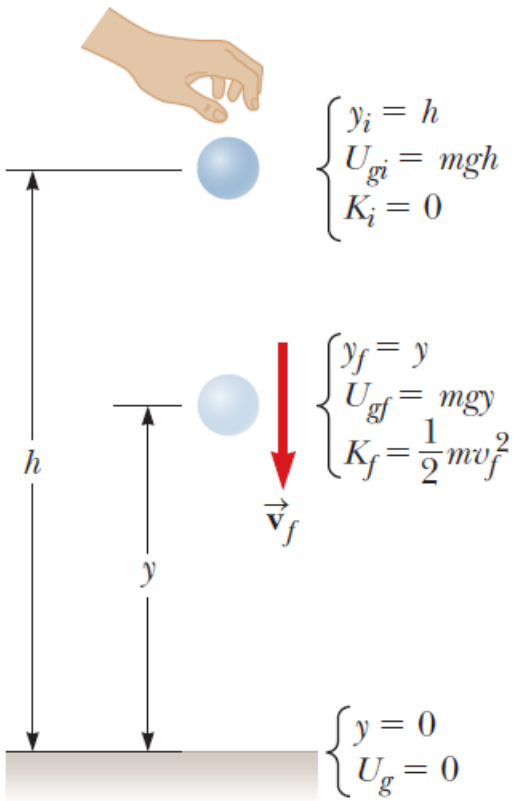
$$\boxed{\Delta E_{system} = \Delta K + \Delta U + \Delta E_{int} = 0}$$

$$\boxed{\Delta E_{mech} = \Delta K + \Delta U \neq 0}$$

Example 8.1 Ball in Free Fall AM

A ball of mass m is dropped from a height h above the ground as shown in Figure 8.4.

(A) Neglecting air resistance, determine the speed of the ball when it is at a height y above the ground. Choose the system as the ball and the Earth.



Solutions

A system of the ball and the Earth: isolated system

Neglecting the air resistance => no nonconservative forces

The conservation of mechanical energy:

$$\Delta E_{mech} = \Delta K + \Delta U = 0$$

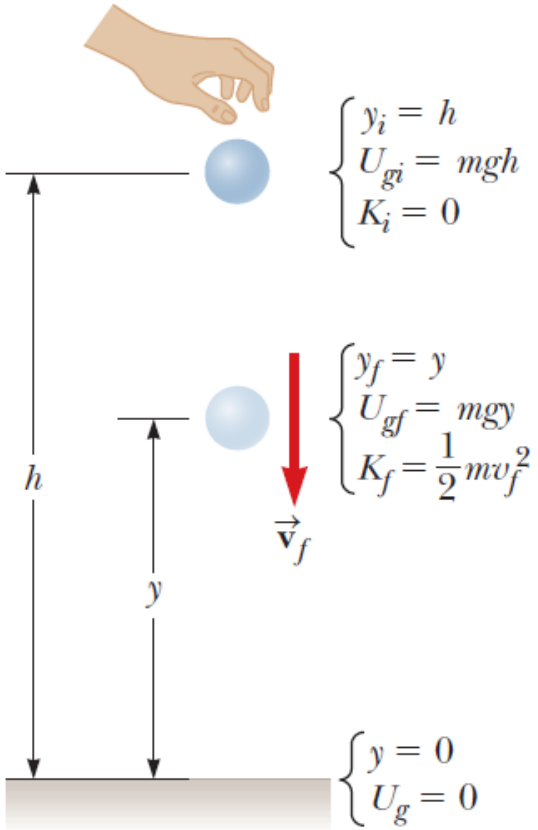
$$\Leftrightarrow \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 + mgy - mgh = 0$$

$$\Leftrightarrow \frac{1}{2}mv_f{}^2 - 0 = mgh - mgy$$

$$\Leftrightarrow \frac{1}{2}mv_f{}^2 - 0 = mgh - mgy$$

$$\Leftrightarrow v_f = \sqrt{2g(h-y)}$$

Gia Bao +



(B) Find the speed of the ball again at height y by choosing the ball as the system.

Solutions:

The work done by the gravitational force on the ball across the boundary of the system

$\Rightarrow$ The system of a ball: nonisolated system

A single object cannot possess potential energy

Huu Khoi +

Energy conservation:

$$\Delta K = W$$

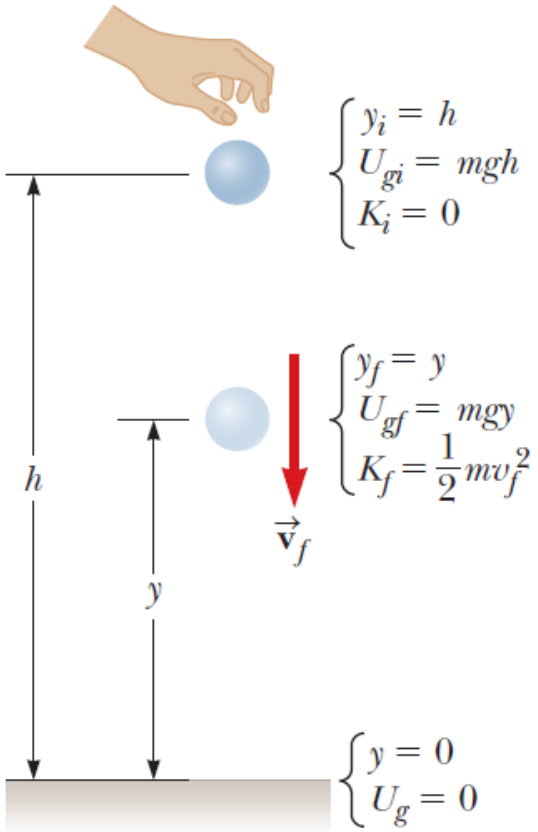
The work done by gravitational force on the ball:

$$W = \vec{F}_g \Delta\vec{r} = mg|\Delta\vec{r}|\cos 0^o = mg(y_i - y_f)$$

$$\Rightarrow \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 = mg(h - y)$$

$$\Leftrightarrow \frac{1}{2}mv_f{}^2 - 0 = \mathrm{mg(h - y)}$$

$$\Leftrightarrow v_f = \sqrt{2g(h - y)}$$



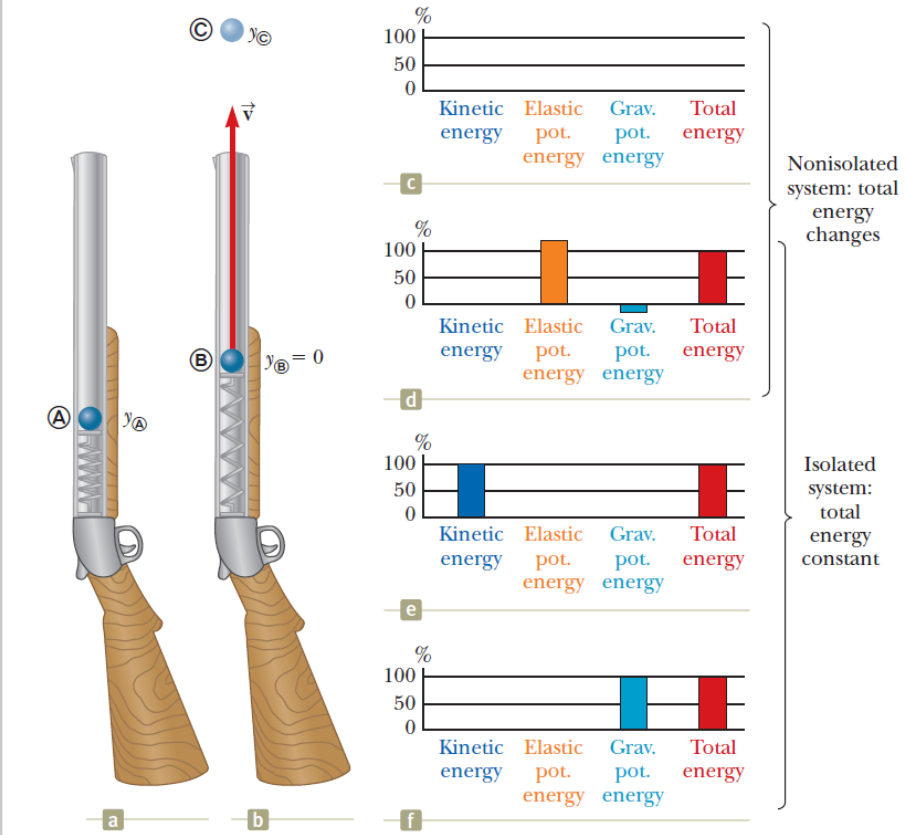
=> The final result is the same, regardless of the choice of system.

Example 8.3 The Spring-Loaded Popgun AM

The launching mechanism of a popgun consists of a trigger-released spring (Fig. 8.6a). The spring is compressed to a position $y_{Ⓐ}$, and the trigger is fired. The projectile of mass m rises to a position $y_{Ⓒ}$ above the position at which it leaves the spring, indicated in Figure 8.6b as position $y_{Ⓑ} = 0$. Consider a firing of the gun for which $m = 35.0$ g, $y_{Ⓐ} = -0.120$ m, and $y_{Ⓒ} = 20.0$ m.

(A) Neglecting all resistive forces, determine the spring constant.

8.3 continued



Solutions

The system of projectile, spring, and the Earth: isolated system.
and no nonconservative force (all resistive forces neglected)
=> Mechanical energy conservation:

$$\Delta E_{mech} = \Delta K + \Delta U = 0$$

$$\Delta E_{mech} = \Delta K + \Delta U = 0$$

$$\Leftrightarrow \Delta K + \Delta U_s + \Delta U_g = 0$$

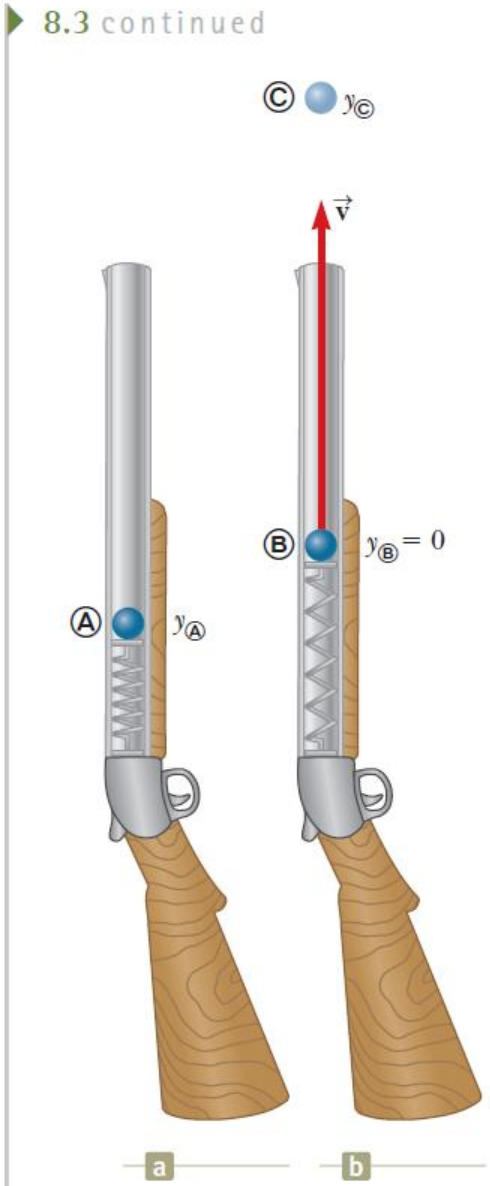
$$\Leftrightarrow \left(\frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2\right) + \left(\frac{1}{2}ky_f{}^2 - \frac{1}{2}ky_i{}^2\right) + (mgy_C - mgy_A) = 0$$

After the gun is fired, the projectile rises to a maximum height at y_C $\Leftrightarrow$ v_f =0

$$\Leftrightarrow (0 - 0) + \left(0 - \frac{1}{2}ky_A{}^2\right) + (mgy_C - mgy_A) = 0$$

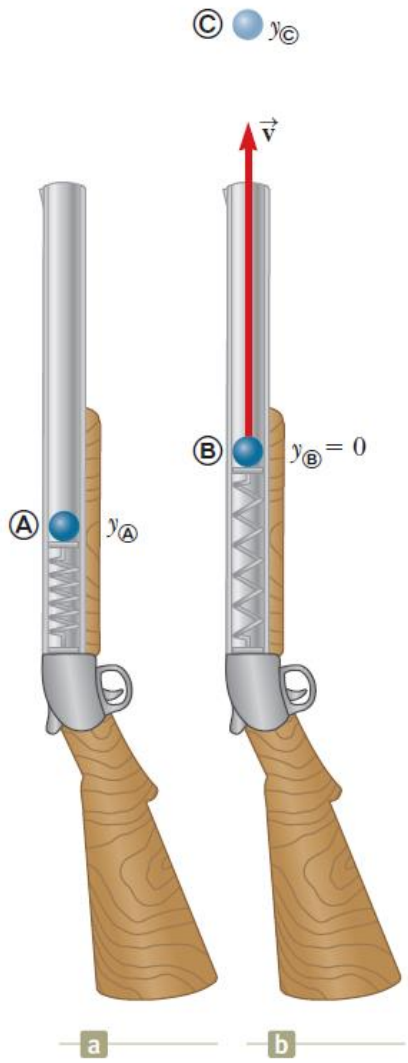
$$\Leftrightarrow \frac{1}{2}ky_A{}^2 = \mathrm{mg}(y_C - y_A)$$

$$\Leftrightarrow k = \frac{2\mathrm{mg}(y_C - y_A)}{y_A{}^2} = \frac{2x0.035x9.8x(20-0.12)}{0.12^2} = 958\,N/m$$



(B) Find the speed of the projectile as it moves through the equilibrium position Ⓑ of the spring as shown in Figure 8.6b.

8.3 continued



Solutions

$$\Delta E_{mech} = \Delta K + \Delta U = 0$$

$$\Leftrightarrow \Delta K + \Delta U_s + \Delta U_g = 0$$

$$\Leftrightarrow \left(\frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2\right) + \left(\frac{1}{2}ky_f{}^2 - \frac{1}{2}ky_i{}^2\right) + (mgy_B - mgy_A) = 0$$

$$\Leftrightarrow \left(\frac{1}{2}mv_B{}^2 - 0\right) + \left(0 - \frac{1}{2}ky_A{}^2\right) + (0 - mgy_A) = 0$$

$$\Leftrightarrow \frac{1}{2}mv_B{}^2 = \frac{1}{2}ky_A{}^2 + mgy_A$$

$$\Leftrightarrow v_B = \sqrt{\frac{ky_A{}^2 + 2mgy_A}{m}} = \sqrt{\frac{958x\ 0.12^2 - 2x0.035x9.8x0.12}{0.035}}$$

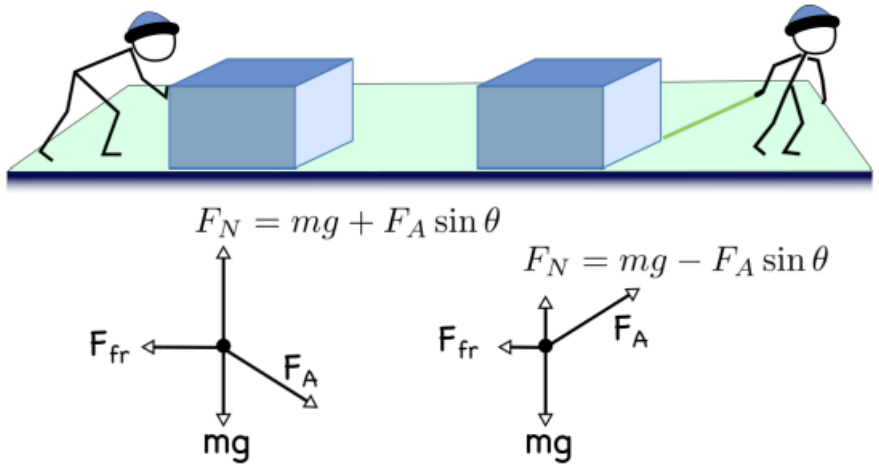
$$= 19.8\ m/s$$

TRƯỜNG HỢP CÓ LỰC MA SÁT

- Lực ma sát trải rộng trên toàn bộ diện tích tiếp xúc

=> Do đó, không thể tính được độ dịch chuyển của điểm đặt mà lực ma sát tác dụng vào.

=> Vì vậy, không thể tính được công thực hiện bởi lực ma sát



TRƯỜNG HỢP CÓ LỰC MA SÁT

Công thực hiện bởi các lực khác lực ma sát:

$$\sum W_{\text{other forces}} = W = \Delta K + \Delta E_{\text{int}}$$

Với $\Delta E_{int} = \mathrm{f_k.d}$

$$\sum W_{\text{other forces}} - f_k d = \Delta K$$

Độ dời

Lực ma sát động

Bảo toàn năng lượng đối với hệ cô lập có lực ma sát tác dụng:

$$\Delta E_{system} = \Delta K + \Delta U + \Delta E_{int} = 0$$

$$\Rightarrow \Delta K + \Delta U + \mathrm{f_k.d} = 0$$

Example 8.4 A Block Pulled on a Rough Surface AM

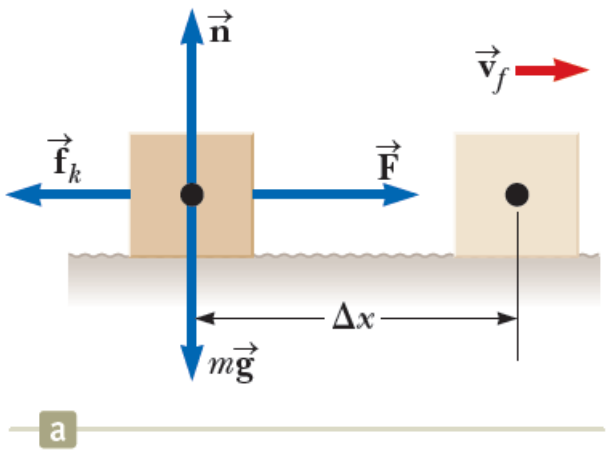
A 6.0-kg block initially at rest is pulled to the right along a horizontal surface by a constant horizontal force of 12 N.

(A) Find the speed of the block after it has moved 3.0 m if the surfaces in contact have a coefficient of kinetic friction of 0.15.

Solutions

The system of the block and surface: nonisolated system.

Work done by other forces than friction force



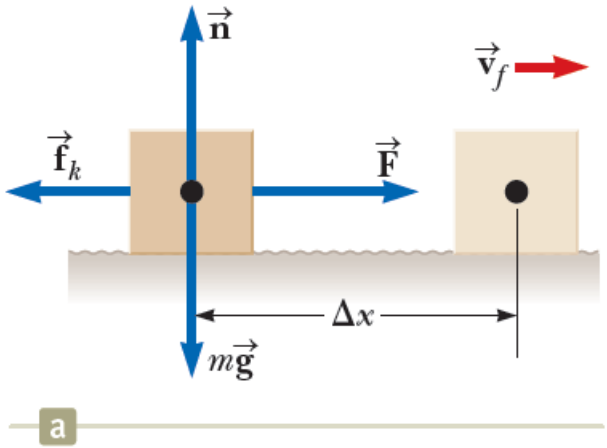
$$\sum W_{other\ forces} = \Delta K + \mathrm{f_k.d}$$

$$\Leftrightarrow \vec{F}\Delta\vec{r} = \Delta K + \mathrm{f_k.d}$$

$$\Leftrightarrow F\Delta x\ cos0^o = \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 + \mathrm{f_k}.\Delta x$$

Solutions

$$F\Delta x\, cos0^o = \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 + \mathrm{f_k}.\Delta x$$



Where $\mathrm{f_k} = \mu_k n$

Apply Newton second's law in y direction:

$$\sum \overrightarrow{F_y} = m\overrightarrow{a_y} \Leftrightarrow \vec{n} + \overrightarrow{F_g} = 0$$

$$\Leftrightarrow \quad n - F_g = 0$$

$$\Leftrightarrow n = mg$$

$$=> \mathrm{f_k} = \mu_k mg$$

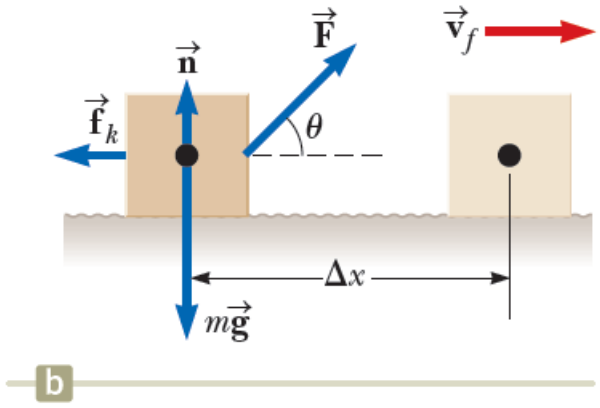
$$\Rightarrow F\Delta x\, cos0^o = \frac{1}{2}mv_f{}^2 - 0 + \mu_k mg\Delta x <=> F\Delta x = \frac{1}{2}mv_f{}^2 + \mu_k mg\Delta x$$

$$\Leftrightarrow v_f = \sqrt{\frac{2(F-\mu_k mg)\Delta x}{m}} = \sqrt{\frac{2(12-0.15x6x9.8)3}{6}} = 1.8 \text{ m/s}$$

(B) Suppose the force $\vec{F}$ is applied at an angle θ as shown in Figure 8.8b. At what angle should the force be applied to achieve the largest possible speed after the block has moved 3.0 m to the right?

Solutions

$$\sum W_{other\ forces} = \Delta K + f_k . d$$



$$\Leftrightarrow \vec{F}\Delta\vec{r} = \Delta K + f_k . d$$

$$\Leftrightarrow F\Delta x\ cos\theta = \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 + f_k . \Delta x$$

Where $f_k = \mu_k n$

Apply Newton second's law in y direction:

$$\sum \overrightarrow{F_y} = m\overrightarrow{a_y} \Leftrightarrow \vec{n} + \overrightarrow{F_g} + \overrightarrow{F_y} = 0$$

$$\Leftrightarrow n - F_g + F_y = 0$$

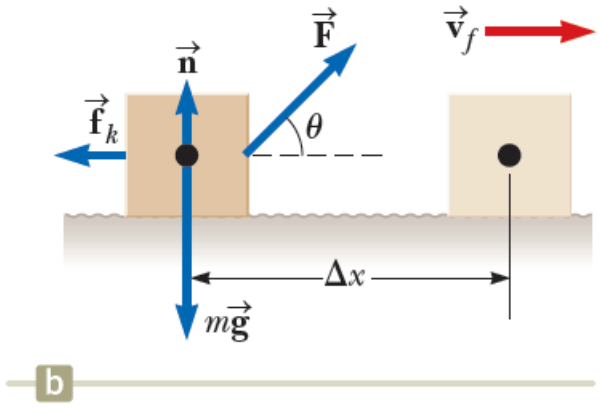
$$\Leftrightarrow n = mg - Fsin\theta$$

=> $f_k = \mu_k(mg - Fsin\theta)$

$$F\Delta x\, cos\theta = \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 + \mathrm{f_k}.\Delta x$$

$$\Leftrightarrow F\Delta x\, cos\theta = \frac{1}{2}mv_f{}^2 + \mu_k(mg - Fsin\theta)\,\Delta x$$

$$\Leftrightarrow \quad v_f = \sqrt{\frac{2(Fcos\theta - \mu_k mg + \mu_k Fsin\theta)\Delta x}{m}}$$



Maximizing the speed

$$\Leftrightarrow \frac{dv_f}{d\theta} = 0$$

$$\Leftrightarrow \frac{1}{2}\sqrt{\frac{2\Delta x}{m}}(-\mathrm{F}\sin\theta + \mu_k \mathrm{F}\cos\theta)(Fcos\theta - \mu_k mg + \mu_k Fsin\theta)^{-\frac{1}{2}} = 0$$

$$\Leftrightarrow -\mathrm{F}\sin\theta + \mu_k \mathrm{F}\cos\theta = 0$$

$$\Leftrightarrow tan\theta = \mu_k = 0.15$$

$$=> \theta = 8.5^o$$

TRƯỜNG HỢP CÓ CÁC LỰC KHÔNG BẢO TOÀN TÁC DỤNG

Pt bảo toàn năng lượng đối với hệ cô lập có tác dụng của lực không bảo toàn:

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

Công thực hiện bởi các lực không bảo toàn đối với hệ không cô lập:

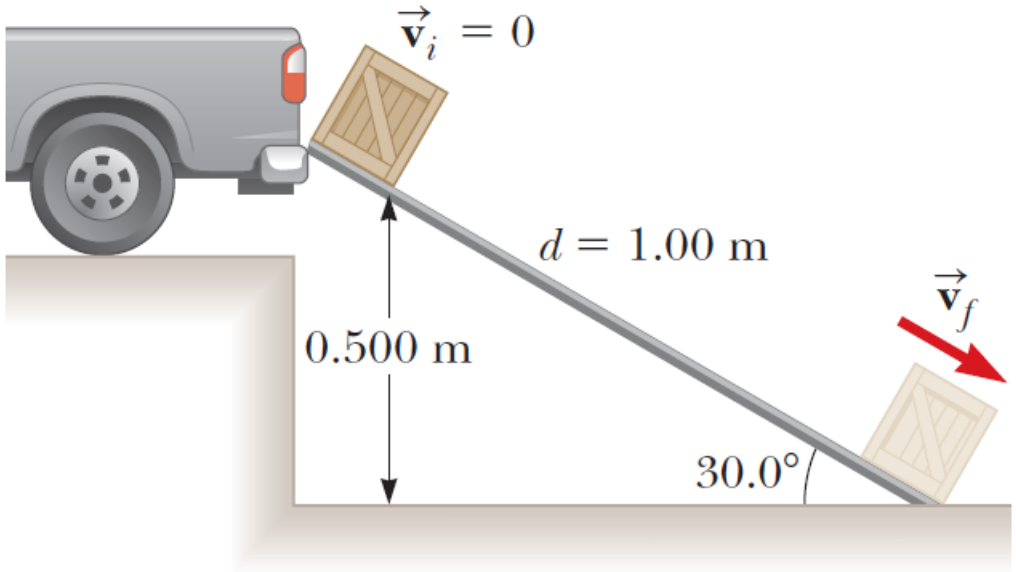
$$\sum W_{\text{other forces}} = W = \Delta K + \Delta U + \Delta E_{\text{int}}$$

Đối với *lực ma sát động*: $\Delta E_{int} = \mathrm{f_k}.\mathrm{d}$

Example 8.7 Crate Sliding Down a Ramp AM

A 3.00-kg crate slides down a ramp. The ramp is 1.00 m in length and inclined at an angle of 30.0° as shown in Figure 8.10. The crate starts from rest at the top, experiences a constant friction force of magnitude 5.00 N, and continues to move a short distance on the horizontal floor after it leaves the ramp.

(A) Use energy methods to determine the speed of the crate at the bottom of the ramp.

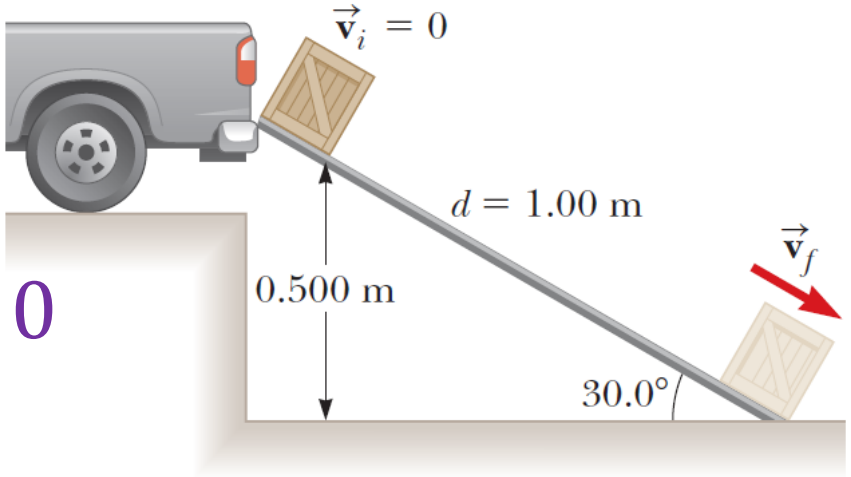


Solutions

The crate, the surface, and the Earth:
isolated system
Conservation of energy:

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$



$$\Leftrightarrow \Delta K + \Delta U + f_k.d = 0$$

$$\Leftrightarrow \left(\frac{1}{2}mv_f^{\,2} - \frac{1}{2}mv_i^{\,2}\right) + (0 - mgh) + f_k.d = 0$$

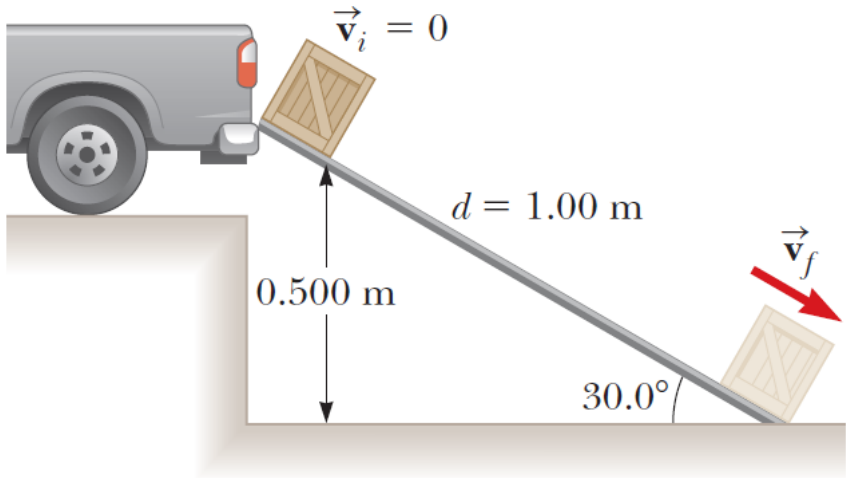
$$\Leftrightarrow \frac{1}{2}mv_f^{\,2} - 0 - mgh + f_k.d = 0$$

$$\Leftrightarrow \frac{1}{2}mv_f^{\,2} = mgh - f_k.d$$

$$\Leftrightarrow v_f = \sqrt{2gh - 2\frac{f_k.d}{m}} = \sqrt{2x9.8x0.5 - 2\frac{5x1}{3}} = 2.54\ m/s$$

(B) How far does the crate slide on the horizontal floor if it continues to experience a friction force of magnitude 5.00 N?

Solutions



Conservation of energy:

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\Leftrightarrow \Delta K + \Delta U + f_k . d = 0$$

$$\Leftrightarrow \left(\frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2\right) + (0 - 0) + f_k . d = 0$$

$$\Leftrightarrow \left(0 - \frac{1}{2} m v_i^2\right) + f_k . d = 0$$

$$\Leftrightarrow d = \frac{1}{2 f_k} m v_i^2 = \frac{1}{2x5} 3x2.54^2 = 1.94\ m$$

Example 8.8 Block–Spring Collision AM

A block having a mass of 0.80 kg is given an initial velocity $v_{Ⓐ} = 1.2$ m/s to the right and collides with a spring whose mass is negligible and whose force constant is $k = 50$ N/m as shown in Figure 8.11.

(A) Assuming the surface to be frictionless, calculate the maximum compression of the spring after the collision.

Solutions

system the block and the spring: *isolated system*

Conservation of mechanical energy:

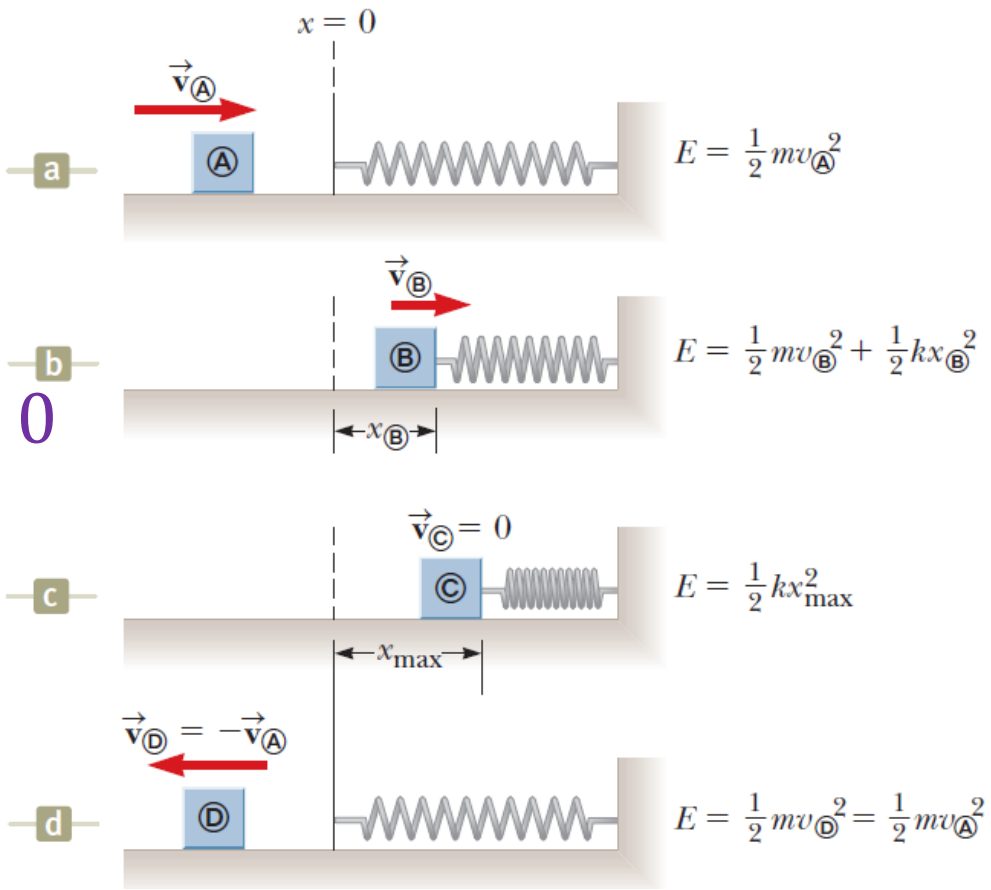
$$\Delta E_{mech} = \Delta K + \Delta U = 0$$

$$\Leftrightarrow \left(\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2\right) + \left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right) = 0$$

$$\Leftrightarrow (0 - \frac{1}{2}mv_A^2) + (\frac{1}{2}kx_{max}^2 - 0) = 0$$

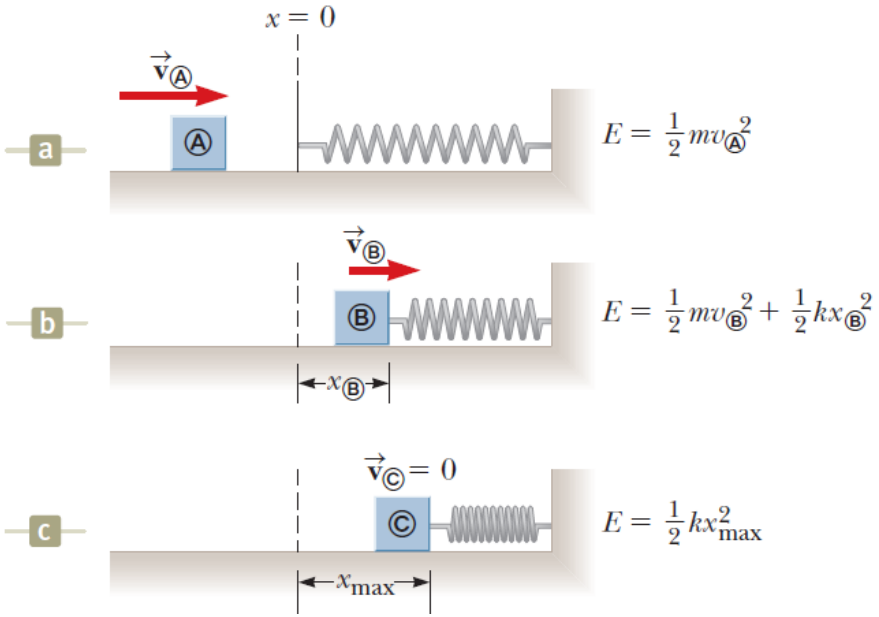
$$\Leftrightarrow \frac{1}{2}mv_A^2 = \frac{1}{2}kx_{max}^2$$

$$\Leftrightarrow x_{max} = \sqrt{\frac{mv_A^2}{k}} = \sqrt{\frac{0.8\text{x}1.2^2}{50}} = 0.15\ m$$



(B) Suppose a constant force of kinetic friction acts between the block and the surface, with $\mu_k = 0.50$. If the speed of the block at the moment it collides with the spring is $v_Ⓐ = 1.2$ m/s, what is the maximum compression $x_Ⓒ$ in the spring?

Solutions



Conservation of energy:

$$\Delta K + \Delta U + \Delta E_{int} = 0$$

$$\Leftrightarrow \Delta K + \Delta U + f_k . d = 0$$

$$\Leftrightarrow \left(\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2\right) + \left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right) + f_k . d = 0$$

$$\Leftrightarrow (0 - \frac{1}{2}mv_A^2) + \left(\frac{1}{2}kx_{max}^2 - 0\right) + \mu_k n x_{max} = 0$$

$$\Leftrightarrow \frac{1}{2}mv_A^2 - \mu_k mg x_{max} = \frac{1}{2}kx_{max}^2$$

$$\Leftrightarrow 50\, x_{max}^2 + 7.84\, x_{max} - 1.15 = 0$$

$$\Leftrightarrow x_{max} = 0.092\, m$$

Example 8.9 Connected Blocks in Motion AM

Two blocks are connected by a light string that passes over a frictionless pulley as shown in Figure 8.12. The block of mass m_1 lies on a horizontal surface and is connected to a spring of force constant k. The system is released from rest when the spring is unstretched. If the hanging block of mass m_2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block of mass m_1 and the surface.

Solutions

System of two blocks, the spring, the surface, and the Earth: isolated system

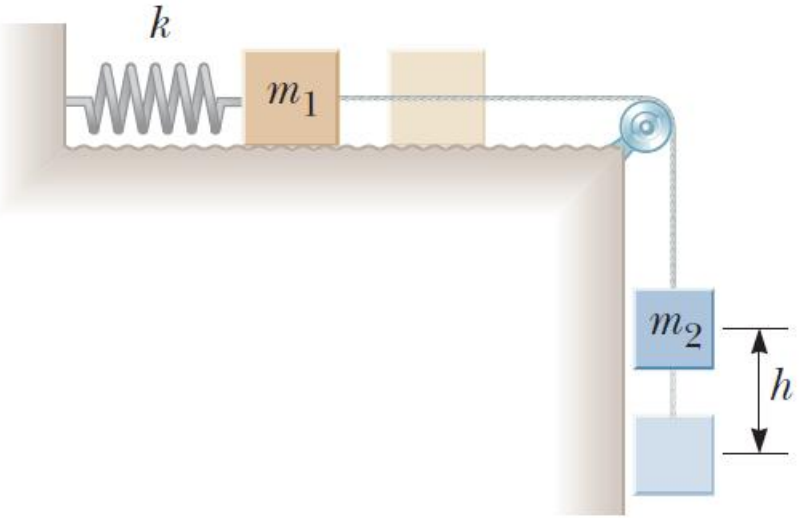
Conservation of energy:

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\Leftrightarrow (0 - 0) + \Delta U_g + \Delta U_s + f_k . d = 0$$

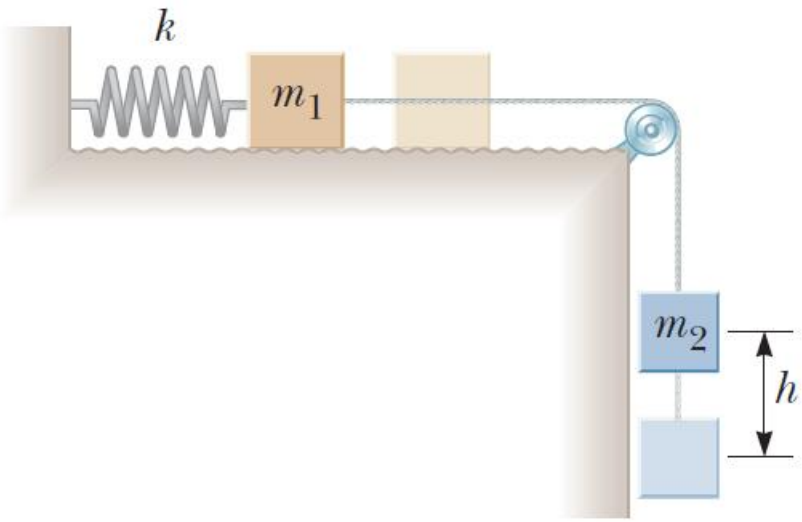
$$\Leftrightarrow (0 - m_2 gh) + \left(\frac{1}{2} k x_f^{\,2} - 0\right) + (\mu_k m_1 g) h = 0$$

$$\Leftrightarrow -m_2 gh + \frac{1}{2} k h^2 + (\mu_k m_1 g) h = 0$$



$$-m_2gh + \frac{1}{2}kh^2 + (\mu_k m_1 g)h = 0$$

$$\Leftrightarrow \mu_k = \frac{m_2}{m_1} - \frac{kh}{m_1 g}$$



Power-Công suất

Công suất tức thời P:

$$P = \frac{dE}{dt}$$

(1W = 1 J/s; 1Hp = 746W)

Truyền năng lượng = công

Công suất trung bình:

$$P_{avg} = \frac{W}{\Delta t}$$

Công suất tức thời P:

$$P = \lim_{\Delta t \to 0} \frac{W}{\Delta t} = \frac{dW}{dt} = \vec{F}.\frac{d\vec{r}}{dt}$$

$$P = \vec{F}.\vec{v}$$

SI unit: J/s

$$1\ \mathrm{W} = 1\ \mathrm{J/s} = 1\ \mathrm{kg \cdot m^2/s^3}$$

HOMEWORK

6. A block of mass $m = 5.00$ kg is released from point Ⓐ **W** and slides on the frictionless track shown in Figure P8.6. Determine (a) the block's speed at points Ⓑ and Ⓒ and (b) the net work done by the gravitational force on the block as it moves from point Ⓐ to point Ⓒ.

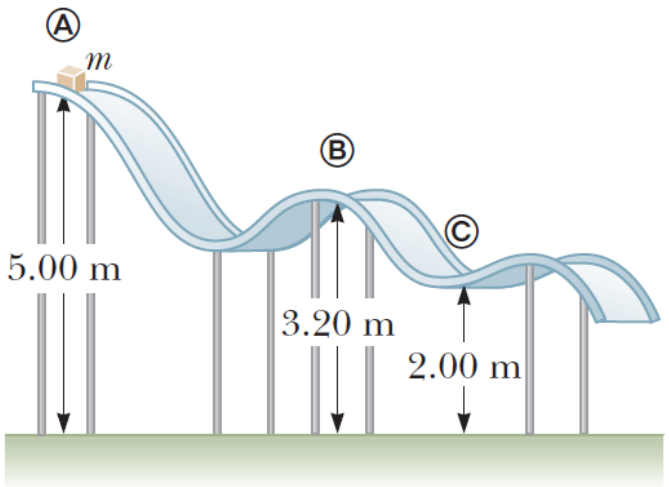


Figure P8.6

7. **M** Two objects are connected by a light string passing over a light, frictionless pulley as shown in Figure P8.7. The object of mass $m_1 = 5.00$ kg is released from rest at a height $h = 4.00$ m above the table. Using the isolated system model, (a) determine the speed of the object of mass $m_2 = 3.00$ kg just as the 5.00-kg object hits the table and (b) find the maximum height above the table to which the 3.00-kg object rises.

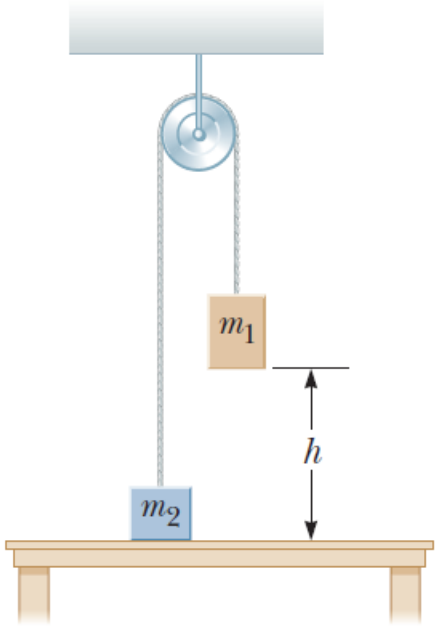


Figure P8.7
Problems 7 and 8.

HOMEWORK

15. A block of mass $m = 2.00$ kg **W** is attached to a spring of force constant $k = 500$ N/m as shown in Figure P8.15. The block is pulled to a position $x_i = 5.00$ cm to the right of equilibrium and released from rest. Find the speed the block has as it passes through equilibrium if (a) the horizontal surface is frictionless and (b) the coefficient of friction between block and surface is $\mu_k = 0.350$.

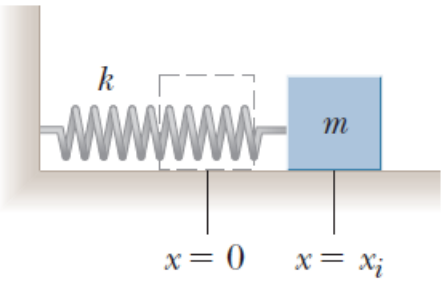


Figure P8.15

22. The coefficient of friction **AMT** between the block of mass **W** $m_1 = 3.00$ kg and the surface in Figure P8.22 is $\mu_k = 0.400$. The system starts from rest. What is the speed of the ball of mass $m_2 = 5.00$ kg when it has fallen a distance $h = 1.50$ m?

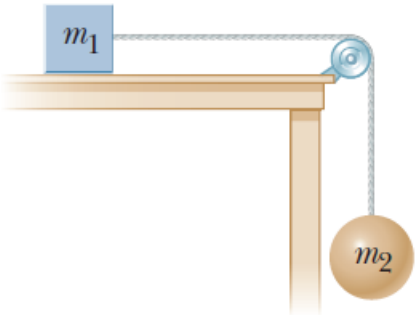


Figure P8.22

23. A 5.00-kg block is set into **M** motion up an inclined plane with an initial speed of $v_i = 8.00$ m/s (Fig. P8.23). The block comes to rest after traveling $d = 3.00$ m along the plane, which is inclined at an angle of $\theta = 30.0°$ to the horizontal. For this motion, determine (a) the change in the block's kinetic energy, (b) the change in the potential energy of the block–Earth system, and (c) the friction force exerted on the block (assumed to be constant). (d) What is the coefficient of kinetic friction?

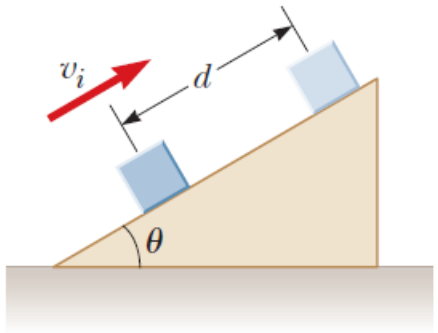


Figure P8.23

HOMEWORK

30. The electric motor of a model train accelerates the train from rest to 0.620 m/s in 21.0 ms. The total mass of the train is 875 g. (a) Find the minimum power delivered to the train by electrical transmission from the metal rails during the acceleration. (b) Why is it the minimum power?

38. A 650-kg elevator starts from rest. It moves upward for 3.00 s with constant acceleration until it reaches its cruising speed of 1.75 m/s. (a) What is the average power of the elevator motor during this time interval? (b) How does this power compare with the motor power when the elevator moves at its cruising speed?

The end ☺